Investigation case 1: SYNTHETIC demo: mid-region reporting gap

This application is an educational maritime analytics portfolio project. Anomaly detections represent statistical or rule-based deviations, not evidence of illegal or hostile activity.

Analysis period: 2030-01-07 04:37 to 2030-01-07 07:39 UTC
Generated: 2026-10-19T18:57:17+00:00

Data sources

• synthetic_scenario / file n/a

Methods

• Detections from configurable rules and an Isolation Forest on vessel-day tracks.
• Evidence attached by an analyst; provenance captured when attached.

Case

Status: Insufficient evidence. Created 2026-10-19 18:57 UTC by local-analyst.
Synthetic vessel stops reporting mid-harbour for ~3 h and reappears further along its route.

Status history

old_status	new_status	changed_by	reason	changed_at
None	New	local-analyst	case opened	2026-10-19 18:57:17.792941+00:00
New	Under review	local-analyst	triage started	2026-10-19 18:57:17.822291+00:00
Under review	Insufficient evidence	local-analyst	Gap alone cannot distinguish coverage lo	2026-10-19 18:57:17.824667+00:00

Detections under review

Event 3920 (reporting_gap): No observations for 3.02 h while the vessel was moving, away from the study-area edge and ports; it moved 5.59 km across the gap. Possible benign explanations: receiver coverage holes, station outages, data-feed gaps, or a powered-down unit. To interpret this, review: coverage maps for the receiver network, satellite AIS, and whether other vessels in the same area also have gaps at that time. This is a rule-based deviation, not evidence of illegal or hostile activity.

Anomaly evidence

anomaly_event_id	anomaly_type	detector	anomaly_mmsi	event_time	score
3920.0	reporting_gap	rule	999010134	2030-01-07 04:37:56+00:00	None

Supporting observations

Positions

position_id	position_mmsi	base_datetime	lat	lon	sog
1705599.0	999010134	2030-01-07 04:37:56+00:00	40.56732465133631	-74.02025721653735	7.0
1705600.0	999010134	2030-01-07 07:39:10+00:00	40.51750296672659	-74.02929728468163	6.7

Analyst notes

Notes

added_at	added_by	note
2026-10-19 18:57:17.819892+00:00	local-analyst	Gap is mid-region (not near the study-ar

Evidence references

• anomaly_events.event_id=3920
• ais_positions.position_id=1705599
• ais_positions.position_id=1705600

Limitations

• AIS is self-reported and received by terrestrial stations; coverage varies and gaps are common. An observation gap is not evidence of deliberate transponder deactivation.
• NOAA Marine Cadastre data is downsampled to roughly one fix per minute per vessel.
• WPI ports are single reference points; 'near port' means within the configured radius of that point, not verified berthing.
• Rule thresholds are heuristics for this study area, not safety or regulatory standards.
• Isolation Forest scores rank statistical unusualness; they are not probabilities of any behaviour and say nothing about intent.